Linear regression + Matching

ECON726

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Housekeeping

Today

- Discuss paper selection
- Finish linear regression
- Introduction to matching
- Review examples in jupyter notebooks (regression; matching)

Homework

-

Empirical project

I use negative grading. I'm sorry it seems harsh! You all had great starts to your projects!

- Limitations of data should be meaningful. Who aren't you looking at?
- Does SUTVA apply?
- Include your paper title, authors etc. Ideally, also include the link! And ideally, in an appendix, include the original figures you're referencing!
- Imagine you are explaining to a coworker who has no idea about the context of your paper.

Empirical project

Paper selection: Due **September 28** [5% of final grade]

- Done!

Project plan: Due **October 13** [15% of final grade]

- Done!

Replication results: Due **November 2**: [10% of final grade]

- You should submit some of your main replication results. The final results should be formatted as tables and charts in a word document or PDF, submitted on Brightspace. Your code and data should accessible on GitHub.

Presentation: Due **December 1** & **December 8** [10% of final grade]

- Prepare a 10 minute presentation on your paper. Focus on the research question, main result, and one extension.

Final Paper: Due **December 11** [40% of final grade]

- Your paper should contain 6-10 pages of text, and 3-6 pages of figures (double spaced, 12 point font). You must provide the code and data to replicate your findings. Your final project should be submitted on Brightspace, and your code and data should be uploaded to github (note: if your data is too large to upload to github, let me know).

Outline

Matching

Control for covariates using the Propensity Score

Propensity scores and treatment effects

Important extensions

Logistic regression

Matching

Control for covariates using the Propensity Score

Propensity scores and treatment effects

Important extensions

Logistic regression

Motivating example

Consider the universe of applicants to the U.S. military:

- Y_i: average earnings
- D_i: dummy variable indicating whether i served in the military

We may be interested in the average effect of treatment on the treated (ToT):

$$\mathbf{E}[Y_i(1) - Y_i(0)|D_i = 1]$$

Difference between:

- Average earnings of soldiers (observable)
- Average earnings of soldiers, had they not served (unobservable)

Naïve comparisons (average predicted effect) will be biased:

$$\mathbf{E}[Y_i|D_i = 1] - \mathbf{E}[Y_i|D_i = 0] = \underbrace{\mathbf{E}[Y_i(1) - Y_i(0)|D_i = 1]}_{\text{ATE on treated}} + \underbrace{\mathbf{E}[Y_i(0)|D_i = 1] - \mathbf{E}[Y_i(0)|D_i = 0]}_{\text{Selection bias}}$$

Motivating example: CIA

Conditional independence assumption:

$$Y_i(0), Y_i(1) \perp D_i | X_i$$

Assuming CIA, selection bias disappears after conditioning on X_i:

$$\delta_{ToT} = \mathbf{E}[Y_i(1) - Y_i(0) | D_i = 1]$$

Motivating example: CIA

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$$\begin{aligned} \delta_{ToT} &= \mathbf{E}[Y_i(1) - Y_i(0) | D_i = 1] \\ &= \mathbf{E}\left[\mathbf{E}[Y_i(1) - Y_i(0) | X_i, D_i = 1] \middle| D_i = 1\right] && \because \text{LIE} \end{aligned}$$

Motivating example: CIA

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&= \mathbf{E}\left[\mathbf{E}[Y_i(1) - Y_i(0) | X_i, D_i = 1] \middle| D_i = 1\right] && \because \text{LIE} \\
&= \mathbf{E}\left[\mathbf{E}[Y_i(1) | X_i, D_i = 1] - \mathbf{E}[Y_i(0) | X_i, D_i = 1] \middle| D_i = 1\right]
\end{aligned}$$

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&= \mathbf{E}\left[\mathbf{E}[Y_i(1) | X_i, D_i = 1] - \mathbf{E}[Y_i(0) | X_i, D_i = 0] \middle| D_i = 1\right]
\end{aligned}$$

By CIA:

$$\mathbf{E}[Y_i(0) | X_i, D_i = 0] = \mathbf{E}[Y_i(0) | X_i, D_i = 1]$$

Motivating example: CIA

Conditional independence assumption:

$$Y_i(0), Y_i(1) \perp D_i | X_i$$

Assuming CIA, selection bias disappears after conditioning on X_i:

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&= \mathbf{E}\left[\mathbf{E}[Y_i(1) | X_i, D_i = 1] - \mathbf{E}[Y_i(0) | X_i, D_i = 1] | D_i = 1\right] \\
&= \mathbf{E}\left[\mathbf{E}[Y_i(1) | X_i, D_i = 1] - \mathbf{E}[Y_i(0) | X_i, D_i = 0] | D_i = 1\right]
\end{aligned}$$

By CIA:

$$\mathbf{E}[Y_i(0) | X_i, D_i = 0] = \mathbf{E}[Y_i(0) | X_i, D_i = 1]$$

By definition of $Y_i = Y_i(0) + D_i(Y_i(1) - Y_i(0))$:

$$\begin{aligned}
\mathbf{E}[Y_i | X_i, D_i = 0] &= \mathbf{E}[Y_i(0) + D_i(Y_i(1) - Y_i(0)) | X_i, D_i = 0] \\
&= \mathbf{E}[Y_i(0) | X_i, D_i = 0] \\
\mathbf{E}[Y_i | X_i, D_i = 1] &= \mathbf{E}[Y_i(1) | X_i, D_i = 1]
\end{aligned}$$

Motivating example: CIA

Conditional independence assumption:

$$Y_i(0), Y_i(1) \perp D_i | X_i$$

Assuming CIA, selection bias disappears after conditioning on X_i:

$$\begin{aligned}
\delta_{ToT} &= \mathbf{E}[Y_i(1) - Y_i(0) | D_i = 1] \\
&= \mathbf{E}\left[\mathbf{E}[Y_i(1) - Y_i(0) | X_i, D_i = 1] | D_i = 1\right] && \because \text{LIE} \\
&= \mathbf{E}\left[\mathbf{E}[Y_i(1) | X_i, D_i = 1] - \mathbf{E}[Y_i(0) | X_i, D_i = 1] | D_i = 1\right] \\
&= \mathbf{E}\left[\mathbf{E}[Y_i(1) | X_i, D_i = 1] - \mathbf{E}[Y_i(0) | X_i, D_i = 0] | D_i = 1\right] \\
&= \mathbf{E}\left[\mathbf{E}[Y_i | X_i, D_i = 1] - \mathbf{E}[Y_i | X_i, D_i = 0] | D_i = 1\right]
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$$\mathbf{E}[Y_i(0) | X_i, D_i = 0] = \mathbf{E}[Y_i(0) | X_i, D_i = 1]$$

By definition of $Y_i = Y_i(0) + D_i(Y_i(1) - Y_i(0))$:

$$\begin{aligned}
\mathbf{E}[Y_i | X_i, D_i = 0] &= \mathbf{E}[Y_i(0) + D_i(Y_i(1) - Y_i(0)) | X_i, D_i = 0] \\
&= \mathbf{E}[Y_i(0) | X_i, D_i = 0] \\
\mathbf{E}[Y_i | X_i, D_i = 1] &= \mathbf{E}[Y_i(1) | X_i, D_i = 1]
\end{aligned}$$

$$\delta_{ToT} = \mathbf{E}\left[\mathbf{E}[Y_i|X_i, D_i = 1] - \mathbf{E}[Y_i|X_i, D_i = 0] \,\middle|\, D_i = 1 \right]$$

Consider the parts of this estimator:

$$\delta_{ToT} = \mathbf{E}\left[\underbrace{\mathbf{E}[Y_i|X_i, D_i = 1]}_{} - \mathbf{E}[Y_i|X_i, D_i = 0] \,\middle|\, D_i = 1 \right]$$

Consider the parts of this estimator:

$\mathbf{E}[Y_i|X_i, D_i = 1]$: **For a given** X_i, take the average over all observed **treated** Y_i

$$\delta_{ToT} = \mathbf{E}\left[\mathbf{E}[Y_i|X_i, D_i = 1] - \underbrace{\mathbf{E}[Y_i|X_i, D_i = 0]}_{} \middle| D_i = 1 \right]$$

Consider the parts of this estimator:

$\mathbf{E}[Y_i|X_i, D_i = 1]$: **For a given** X_i, take the average over all observed **treated** Y_i

$\mathbf{E}[Y_i|X_i, D_i = 0]$: **For a given** X_i, take the average over all observed **untreated** Y_i

$$\delta_{ToT} = \mathbf{E}\left[\underbrace{\mathbf{E}[Y_i|X_i, D_i = 1] - \mathbf{E}[Y_i|X_i, D_i = 0]}\Big| D_i = 1\right]$$

Consider the parts of this estimator:

$\mathbf{E}[Y_i|X_i, D_i = 1]$: **For a given** X_i, take the average over all observed **treated** Y_i

$\mathbf{E}[Y_i|X_i, D_i = 0]$: **For a given** X_i, take the average over all observed **untreated** Y_i

Define δ_X to be the average difference in oberved Y_i for treated and untreated with a given value X_i:

$$\delta_X \equiv \underbrace{\mathbf{E}[Y_i|X_i, D_i = 1] - \mathbf{E}[Y_i|X_i, D_i = 0]}$$

- Example: difference in mean earnings by veteran status at each value of X_i

$$\delta_{ToT} = \mathbf{E}\left[\mathbf{E}[Y_i|X_i, D_i = 1] - \mathbf{E}[Y_i|X_i, D_i = 0] \,\middle|\, D_i = 1 \right]$$

$$= \mathbf{E}\left[\delta_X \,\middle|\, D_i = 1\right]$$

Consider the parts of this estimator:

$\mathbf{E}[Y_i|X_i, D_i = 1]$: **For a given** X_i, take the average over all observed **treated** Y_i

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- Example: difference in mean earnings by veteran status at each value of X_i

Effect of treatment on the treated under CIA:

$$\begin{aligned}\delta_{ToT} &= \mathbf{E}\left[\mathbf{E}[Y_i|X_i, D_i = 1] - \mathbf{E}[Y_i|X_i, D_i = 0]|\, D_i = 1\right] \\ &= \mathbf{E}\left[\delta_X|\, D_i = 1\right]\end{aligned}$$

where

$$\delta_X = \mathbf{E}[Y_i|X_i, D_i = 1] - \mathbf{E}[Y_i|X_i, D_i = 0]$$

Let's say there was some variable X_i that represents an indicator of whether an applicant is "high ability" or "lower ability"

- $X_i = 0$ means i is low ability
- $X_i = 1$ means i is high ability

We'll assume CIA: $\{Y_i(0), Y_i(1)\} \perp D_i|X_i$

- Intuitively, what does this mean? Is this a good assumption? Is it realistic?

The matching estimand can be written as:

$$\begin{aligned}\delta_{ToT} = \mathbf{E}[Y_i(1) - Y_i(0)|D_i = 1] &= \mathbf{E}[\delta_X|D_i = 1] \\ &= \sum_{x\in\{0,1\}} \delta_x \mathbf{P}(X_i = x|D_i = 1) \\ &= \delta_0 \mathbf{P}(X_i = 0|D_i = 1) + \delta_1 \mathbf{P}(X_i = 1|D_i = 1)\end{aligned}$$

$$\begin{aligned}\delta_{ToT} &= \mathbf{E}[\delta_X | D_i = 1] \\ &= \delta_{X_i=0}\mathbf{P}(X_i = 0|D_i = 1) + \delta_{X_i=1}\mathbf{P}(X_i = 1|D_i = 1)\end{aligned}$$

How do these elements connect to data?

$$\begin{aligned}\delta_{ToT} &= \mathbf{E}[\delta_X | D_i = 1] \\ &= \delta_{X_i=0}\mathbf{P}(X_i = 0|D_i = 1) + \delta_{X_i=1}\mathbf{P}(X_i = 1|D_i = 1)\end{aligned}$$

How do these elements connect to data?

$\delta_{X_i=0}$ Difference in observed Y_i for treated and untreated with $X_i = 0$
(Average difference in wages by veteran status for **low** ability i)

$$\begin{aligned}\delta_{ToT} &= \mathbf{E}[\delta_X | D_i = 1] \\ &= \delta_{X_i=0}\mathbf{P}(X_i = 0|D_i = 1) + \delta_{X_i=1}\mathbf{P}(X_i = 1|D_i = 1)\end{aligned}$$

How do these elements connect to data?

$\delta_{X_i=0}$ Difference in observed Y_i for treated and untreated with $X_i = 0$
(Average difference in wages by veteran status for **low** ability i)

$\delta_{X_i=1}$ Difference in observed Y_i for treated and untreated with $X_i = 1$
(Average difference in wages by veteran status for **high** ability i)

$$\begin{aligned}\delta_{ToT} &= \mathbf{E}[\delta_X | D_i = 1] \\ &= \delta_{X_i=0}\mathbf{P}(X_i = 0|D_i = 1) + \delta_{X_i=1}\mathbf{P}(X_i = 1|D_i = 1)\end{aligned}$$

How do these elements connect to data?

$\delta_{X_i=0}$	Difference in observed Y_i for treated and untreated with $X_i = 0$ (Average difference in wages by veteran status for **low** ability i)
$\delta_{X_i=1}$	Difference in observed Y_i for treated and untreated with $X_i = 1$ (Average difference in wages by veteran status for **high** ability i)
$\mathbf{P}(X_i = 0\|D_i = 1)$	Probability $X_i = 0$ conditional on being treated (Fraction of veterans that are **low** ability)

$$\begin{aligned}\delta_{ToT} &= \mathbf{E}[\delta_X | D_i = 1] \\ &= \delta_{X_i=0}\mathbf{P}(X_i = 0|D_i = 1) + \delta_{X_i=1}\mathbf{P}(X_i = 1|D_i = 1)\end{aligned}$$

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$\mathbf{P}(X_i = 1\|D_i = 1)$	Probability $X_i = 1$ conditional on being treated (Fraction of veterans that are **high** ability)

$$\begin{aligned}\delta_{ToT} &= \mathbf{E}[\delta_X | D_i = 1] \\ &= \delta_{X_i=0}\mathbf{P}(X_i = 0|D_i = 1) + \delta_{X_i=1}\mathbf{P}(X_i = 1|D_i = 1)\end{aligned}$$

How do these elements connect to data?

$\delta_{X_i=0}$	Difference in observed Y_i for treated and untreated with $X_i = 0$ (Average difference in wages by veteran status for **low** ability i)
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$\mathbf{P}(X_i = 0\|D_i = 1)$	Probability $X_i = 0$ conditional on being treated (Fraction of veterans that are **low** ability)
$\mathbf{P}(X_i = 1\|D_i = 1)$	Probability $X_i = 1$ conditional on being treated (Fraction of veterans that are **high** ability)

Intuitively, what is this matching estimate?

- X_i-specific treatment-control comparisons, weighted together to produce a single overall treatment effect

Effect of treatment on the treated under CIA:

$$\begin{aligned}\delta_{ToT} &= \mathbf{E}\left[\mathbf{E}[Y_i|X_i, D_i = 1] - \mathbf{E}[Y_i|X_i, D_i = 0]|\, D_i = 1\right] \\ &= \mathbf{E}\left[\delta_X|\, D_i = 1\right]\end{aligned}$$

where

$$\delta_X = \mathbf{E}[Y_i|X_i, D_i = 1] - \mathbf{E}[Y_i|X_i, D_i = 0]$$

Now instead of X_i being a dummy, assume X_i is the result of test of ability/intelligence

- $X_i \in \{0, 1, \ldots, 100\}$ and measures ability

We'll assume CIA: $\{Y_i(0), Y_i(1)\} \perp D_i|X_i$

- Intuitively, what does this mean? Is this a good assumption? Is it realistic?

The matching estimand can be written as:

$$\begin{aligned}\delta_{ToT} &= \mathbf{E}[\delta_X|D_i = 1] \\ &= \sum_{x=0}^{100} \delta_x \mathbf{P}(X_i = x|D_i = 1) \\ &= \delta_0 \mathbf{P}(X_i = 0|D_i = 1) + \delta_1 \mathbf{P}(X_i = 1|D_i = 1) + \cdots + \delta_{100} \mathbf{P}(X_i = 100|D_i = 1)\end{aligned}$$

Intuitively, what is this matching estimate?

- X_i-specific treatment-control comparisons, weighted together to produce a single overall treatment effect

Now instead of X_i isn't a single variable, but instead a vector of (discrete) variables

- Year of birth, test score, year of application to the military, educational attainment at the time of application

Think of each realized $X_i = x$ as a particular combination of all these variables

- For example, $X_i = z$ might describe an applicant born in 1970, who scored a 95 on an ability test, and graduated from college

We'll assume CIA: $\{Y_i(0), Y_i(1)\} \perp D_i | X_i$

- Intuitively, what does this mean? Is this a good assumption? Is it realistic?

The matching estimand can be written as:

$$\begin{aligned}\delta_{ToT} &= \mathbf{E}[\delta_X | D_i = 1] \\ &= \sum_x \delta_x \mathbf{P}(X_i = x | D_i = 1)\end{aligned}$$

Intuitively, what is this matching estimate?

- The δ_x are x-specific treatment-control comparisons
  - For $X_i = z$, compare average wage difference of veterans and non-veterans who were born in 1970, who scored a 95 on an ability test, and graduated from college
  - Difference in average wages by veteran status for those with the same X_i
- Weighted together to produce a single overall treatment effect
  - Calculate weights based on the distribution of characteristics X_i among veterans

Unconditional Average Treatment Effect

Conditional treatment effect on treated:

$$\begin{aligned}\delta_{ToT} = \mathbf{E}[Y_i(1) - Y_i(0)|D_i = 1] &= \mathbf{E}[\delta_X|D_i = 1] \\ &= \sum_{x\in\{0,1\}} \delta_x \mathbf{P}(X_i = x|D_i = 1) \\ &= \delta_0 \mathbf{P}(X_i = 0|D_i = 1) + \delta_1 \mathbf{P}(X_i = 1|D_i = 1)\end{aligned}$$

- How much the typical soldier gained or lost as a result of enlisting

Can also write unconditional average treatment effect:

$$\begin{aligned}\delta_{ATE} &= \mathbf{E}\left[\mathbf{E}[Y_i(1)|X_i, D_i = 1] - \mathbf{E}[Y_i(0)|X_i, D_i = 0]\right] \\ &= \mathbf{E}[Y_i(1) - Y_i(0)] \\ &= \sum_x \delta_x \mathbf{P}(X_i = x)\end{aligned}$$

- How much the typical applicant would gain or lose as a result of enlisting

Matching and regression

Matching and regression are related:

- Both are comparing outcomes (differences in Y_i) over treatment groups, controlling for covariates X_i
- These methods differ in implementation

Regression estimates of matching strategy:

$$Y_i = \sum_x d_{ix}\alpha_x + \delta_R D_i + e_i$$

where:

- $d_{ix} = \mathbf{1}[X_i = x]$: dummy variable that equals 1 if $X_i = x$, 0 otherwise
- α_x: regression effect for $X_i = x$
- δ_R: regression estimand

Matching and regression differences

In general, estimates from matching and from regression will often be similar, but not identical.

- Regression can be thought of as a type of matching strategy
  - Average effects are calculated by covariates X: δ_X
  - These effects are weighed to produce a weighted average
- Difference in how weights are determined. For ATE on treated:
  - Matching: weights are determined by distribution of covariates on treated
  - Regression: uses treatment-variance weighted average
  - These will differ unless treatment is independent of covariates

Note: neither regression nor covariate matching estimands give any weights to covariate cells that do not contain treated and control observations

- Often called **common support**

Matching

Control for covariates using the Propensity Score

Propensity scores and treatment effects

Important extensions

Logistic regression

Review: matching

Variables:

- Y_i: Outcome variable
- D_i: Binary treatment dummy variable
- X_i: Covariates

Under conditional independence assumption (CIA) $\{Y_i(0), Y_i(1)\} \perp D_i|X_i$, we can construct causal matching estimators:

$$\begin{aligned}\delta_{ToT} &= \mathbf{E}[Y_i(1) - Y_i(0)|D_i = 1] \\ &= \sum_x \delta_x \mathbf{P}(X_i = x|D_i = 1) \\ \delta_{ATE} &= \mathbf{E}[Y_i(1) - Y_i(0)] \\ &= \sum_x \delta x \mathbf{P}(X_i = x)\end{aligned}$$

where:

$$\delta_X = \mathbf{E}[Y_i|X_i, D_i = 1] - \mathbf{E}[Y_i|X_i, D_i = 0]$$

Note: matching and regression are both weighted averages of covariate specific differences

Motivating example

"Growth mindset" example (heavily borrows from: link)

- If you think your intelligence is fixed, there is no incentive to study; academic outcomes are worse
- If you have a "growth mindset," there are returns to intelligence; your academic outcomes are better

The National Study of Learning Mindsets

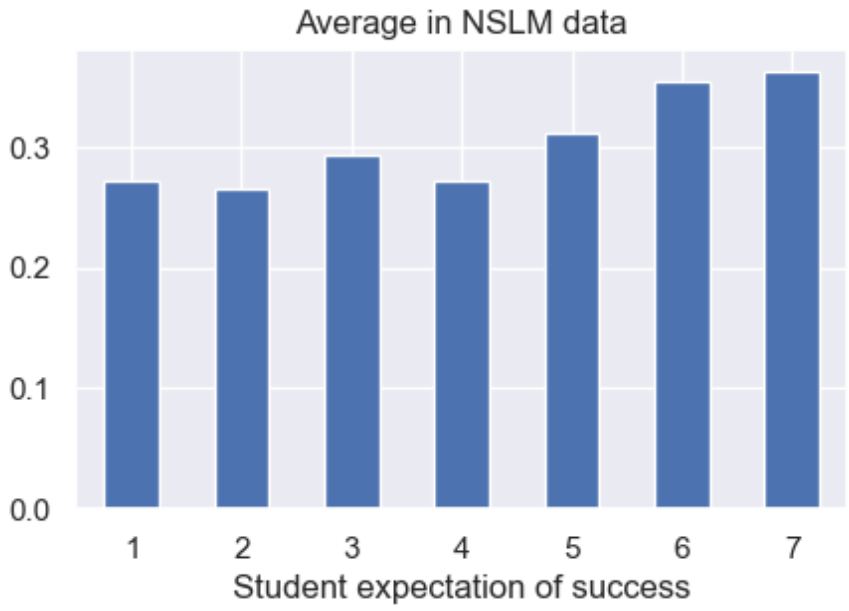
- Evaluate the impact of a nudge-like intervention designed to instill students with a growth mindset on student achievement
- Students randomly given the opportunity to participate in "growth mindset" seminar in high school; academic outcomes in college were then recorded
- Problem: students with the highest expectation of success were more likely to receive the treatment
  - Opportunity to participate in the seminar was random
  - Those who moved forward with the opportunity may have been more ambitious than those who did not

Athey and Wagner (2016) generate simulated data: link

Motivating example (cont)

Variables:

- Y_i: Outcome variable (achievement score in college)
- D_i: Binary treatment dummy variable (participating in "growth mindset") seminar
- X_i: Covariates (individual and school demographics; family characteristics; survey about expectations of future success)



More ambitious students may be more willing to go to the seminar, leading to selection bias. How can we correct for that?

Propensity Score Theorem

Define the propensity score (or balancing score) to be:

$$\begin{aligned} p(X_i) =& \mathbf{E}[D_i|X_i] \\ =& \sum_{d=0}^{1} d \cdot \mathbf{P}(D_i = d|X_i) \\ =& 0 \cdot \mathbf{P}(D_i = 0|X_i) + 1 \cdot \mathbf{P}(D_i = 1|X_i) \\ =& \mathbf{P}(D_i = 1|X_i) \end{aligned}$$

Note: the propensity score equals the probability of treatment for binary treatments

Suppose the CIA holds such that $\{Y_i(0), Y_i(1)\} \perp D_i|X_i$. Then $\{Y_i(0), Y_i(1)\} \perp D_i|p(X_i)$

What does this mean?

- You don't need to directly control for confounders X to achieve CIA
- Instead, it is sufficient to control for a single variable, $p(X_i)$

Propensity Score Theorem: Given propensity score:

$$p(X_i) = \mathbf{E}[D_i|X_i] = \mathbf{P}(D_i = 1|X_i)$$

Suppose the CIA holds such that $\{Y_i(0), Y_i(1)\} \perp D_i|X_i$. Then $\{Y_i(0), Y_i(1)\} \perp D_i|p(X_i)$

Proof:
First, instead of writing $\{Y_i(0), Y_i(1)\}$, we'll write $Y_i(j)$, where $j \in 0, 1$.

Next, note that the conditional expectation of the treatment variable equals the conditional probability of treatment (because the treatment is binary):

$$\begin{aligned}
\mathbf{E}[D_i|Y_i(j), p(X_i)] &= \sum_{d=0}^{1} d\mathbf{P}(D_i = d|Y_i(j), p(X_i)) \\
&= 0 \cdot \mathbf{P}(D_i = 0|Y_i(j), p(X_i)) + 1 \cdot \mathbf{P}(D_i = 1|Y_i(j), p(X_i)) \\
&= \mathbf{P}(D_i = 1|Y_i(j), p(X_i))
\end{aligned}$$

Propensity Score Theorem: Given propensity score:

$$p(X_i) = \mathbf{E}[D_i|X_i] = \mathbf{P}(D_i = 1|X_i)$$

Suppose the CIA holds such that $\{Y_i(0), Y_i(1)\} \perp D_i|X_i$. Then $\{Y_i(0), Y_i(1)\} \perp D_i|p(X_i)$

Proof:
We want to show that:

$$\mathbf{E}[D_i|X_i, p(X_i)] = \mathbf{E}[D_i|p(X_i)]$$

Propensity Score Theorem: Given propensity score:

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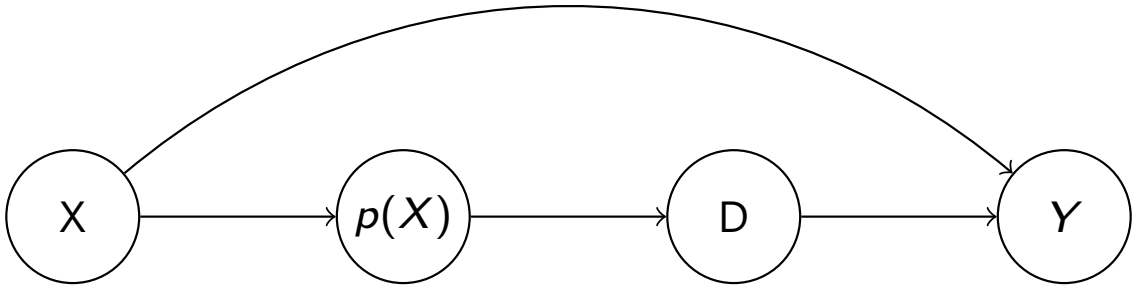
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Because $p(X_i) = p(X_i)$, the statement is true.



- If I know what $p(X)$ is, then X alone doesn't tell me anything else about what the probability of treatment is
- Controlling for $p(X)$ acts the same way as controlling for X directly

Matching

Control for covariates using the Propensity Score

Propensity scores and treatment effects

Important extensions

Logistic regression

Propensity scores and treatment effects

How are propensity scores used in practice?

- First, estimate the propensity score
  - Use a parametric model, like a logit or probit model
- Next generate the effect of treatment
  - Match on estimated score (direct propensity score matching)
  - OR use weighting scheme described next

Direct propensity score matching

Given a propensity score, $p(X_i)$, and the CIA $\{Y_i(0), Y_i(1)\} \perp D_i|X_i$:

$$\begin{aligned}\delta_{ToT} =& \mathbf{E}[Y_i(1) - Y_i(0)|D_i = 1] \\ =& \mathbf{E}\left[\mathbf{E}[Y_i|p(X_i), D_i = 1] - \mathbf{E}[Y_i|p(X_i), D_i = 0]\right]\end{aligned}$$

Just like we did for matching, you can

- Stratify the sample in to values of $p(X_i)$
- Find treatment effects conditional on similar $p(X_i)$
- Weigh these treatment effects by the distribution of $p(X_i)$ conditional on treatment

Propensity score weights

Alternative: estimate treatment effects using propensity score weights. Final equation:

$$\mathbf{E}[Y_i(1) - Y_i(0)] = \mathbf{E}\left[\frac{Y_i\left[D_i - p(X_i)\right]}{p(X_i)(1 - p(X_i))}\right]$$

We'll go through the math. Motivation for this method:

- Avoids the matching step
- The concept of inverse probability weighting is important in more advanced topics

To prove this, we'll have to prove a few interim steps

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Using the above equality, the conditional independence assumption, and the definition of the propensity score:

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Proof: Recall $Y_i = Y_i(0) + (Y_i(1) - Y_i(0))$, and note $D_i(1-D_i) = 0$ for $D_i = 0$ and $D_i = 1$:

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=& Y_i(0)(1-D_i) + (Y_i(1) - Y_i(0))D_i(1-D_i) \\
=& Y_i(0)(1-D_i) + (Y_i(1) - Y_i(0)) \cdot 0 \\
=& Y_i(0)(1-D_i)
\end{aligned}$$

Using the above equality, the conditional independence assumption, and the definition of the propensity score:

$$\begin{aligned}
\mathbf{E}[Y_i(1-D_i)|X_i] =& \mathbf{E}\left[\, Y_i(0)(1-D_i) |\, X_i\right] \\
=& \mathbf{E}[Y_i(0)|X_i] \cdot \mathbf{E}[1-D_i|X_i] && \because\ Y_i(0) \perp D_i |\, X_i \\
=& \mathbf{E}[Y_i(0)|X_i](1-p(X_i))
\end{aligned}$$

Fact #2:

$$\mathbf{E}\left[\frac{Y_i(1-D_i)}{1-p(X_i)}\right] = \mathbf{E}[Y_i(0)]$$

Proof (cont.):

We've seen:

$$\mathbf{E}[Y_i(1-D_i)|X_i] = \mathbf{E}[Y_i(0)|X_i](1-p(X_i))$$

Now we can estimate $\mathbf{E}\left[\frac{Y_i(1-D_i)}{1-p(X_i)}\right]$:

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$$\mathbf{E}\left[\frac{Y_i D_i}{p(X_i)}\right] = \mathbf{E}[Y_i(1)], \qquad \mathbf{E}\left[\frac{Y_i(1-D_i)}{1-p(X_i)}\right] = \mathbf{E}[Y_i(0)]$$

Average treatment effects can be estimated from:

$$\begin{aligned}
\mathbf{E}[Y_i(1) - Y_i(0)] =& \mathbf{E}\left[\frac{Y_i D_i}{p(X_i)} - \frac{Y_i(1-D_i)}{1-p(X_i)}\right] \\
=& \mathbf{E}\left[\frac{Y_i D_i(1-p(X_i)) - Y_i(1-D_i)p(X_i)}{p(X_i)(1-p(X_i))}\right] \\
=& \mathbf{E}\left[\frac{Y_i\left[D_i(1-p(X_i)) - (1-D_i)p(X_i)\right]}{p(X_i)(1-p(X_i))}\right] \\
=& \mathbf{E}\left[\frac{Y_i\left[D_i - D_i p(X_i) - p(X_i) + D_i p(X_i)\right]}{p(X_i)(1-p(X_i))}\right] \\
=& \mathbf{E}\left[\frac{Y_i\left[D_i - p(X_i)\right]}{p(X_i)(1-p(X_i))}\right]
\end{aligned}$$

Horvitz-Thompson Propensity score weighting:

$$\mathbf{E}[Y_i(1) - Y_i(0)] = \mathbf{E}\left[\frac{Y_i\left[D_i - p(X_i)\right]}{p(X_i)(1 - p(X_i))}\right]$$

- Uses propensity score weighting to recover average treatment effects
- Using the propensity score rather than identification by conditioning on X is a useful empirical strategy when X is high-dimensional and $p(X)$ is available (or can be approximated accurately)
- MHE shows how this can be considered similar to the regression estimator

Matching

Control for covariates using the Propensity Score

Propensity scores and treatment effects

Important extensions

Logistic regression

Inverse Probability of Treatment Weighting

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Consider the first element:

- $Y_i D_i$: Isolates Y_i that are treated
- $p(X_i)$: Inverse probability of being treated
  - Probabilities are between 0 and 1
  - Low probability events $\implies$ inverse probability high

$$p(X) = 0.1 = \frac{1}{10} \implies \frac{1}{p(X)} = \frac{1}{1/10} = 10$$

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The same reasoning applies to the second element

IPTW: valuable in a clinical setting to generate balanced data

Structural model and connection to machine learning

Note: the following is from *Business Data Science* by Matt Taddy, Chapter 6

Consider a common structural model:

$$Y = D\gamma + X'\beta + \epsilon, \quad \epsilon|D, X = 0$$
$$D = X'\tau + v, \quad v|X = 0$$

Note:

- $\epsilon|D, X = 0$ and $v|X = 0$ encode CIA

Two-step **linear treatment effects (LTE)** model

- First estimate treatment D_i
- Then estimate treatment effects γ

Dimensionality starts to matter here

- If n (number of observations) are much greater than p (number of X variables), you can use regression models
- If you don't have $n >> p$, traditional methods break down

Estimation of LTE

Structural model:

$$Y = D\gamma + X'\beta + \epsilon, \quad \epsilon|D, X = 0$$
$$D = X'\tau + v, \quad v|X = 0$$

Estimation procedure:

1. Estimate $\mathbf{E}[D|X] = X'\tau$. Collect fitted values $\hat{D} = X'\hat{\tau}$
2. Estimate:
$$\mathbf{E}[Y|X, D] = \hat{D}\psi + D\gamma + X'\beta$$

The ψ is the treatment effect of D on Y.

LTE and propensity score matching

Structural model in propensity score models:

$$Y = D\gamma + X'\beta + \epsilon, \quad \epsilon | D, X = 0$$
$$D \sim \text{Bernoulli}\left(p(X)\right)$$
$$p(X) = \frac{e^{X'\tau}}{1 + e^{X'\tau}}$$

- D is modeled from a Bernoulli distribution (i.e. coin toss) with probability p
- The probability that $D = 1$ is modeled using a **logistic regression**

Once you estimate the propensity score $p(X)$, you can find treatment effects by:

1. Fitting the linear regression: $\mathbf{E}[Y|D, \hat{p}(X)] = \alpha + D\gamma + \hat{p}(X)\psi$
2. Constructing propensity score weighting estimates: $\mathbf{E}\left[\left.\frac{Y}{p(\hat{X})}\right| D\right] = \alpha + D\gamma$
3. Matching

Hahn (1998) shows there is a cost to propensity score matching in terms of asymptotic efficiency

- What is the advantage of, say, controlling for propensity score instead of controlling for covariates in a regression?

MHE : "An essential element of propensity score methods is the use of prior knowledge for dimension reduction. The statistical payoff is an improvement in finite sample behavior. If you're not prepared to smooth, restrict, or otherwise reduce the dimensionality of the matching problem in a manner that has real empirical consequences, the you might as well go for full covariate matching or saturated regression control."

- This is very prescient! Now, in the era of big data, dimensionality reduction is key
- Propensity scores have a new value in double-robust machine learning; see chapter 9 of `https://chapters.causalml-book.org/CausalML_chap_9.pdf`

Matching

Control for covariates using the Propensity Score

Propensity scores and treatment effects

Important extensions

Logistic regression

Why logistic regression

- We have to use some estimation procedure to predict propensity scores $p(X)$
- Logistic regression is a common method
- Quick review of logistic regression and generalized linear models

Linear regression

Assume we have data $\{y_i, x_{1i}, x_{2i}, \ldots, x_{ki}\} = \{y_i, \mathbf{x}_i\}$. A linear regression model takes the form:

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \cdots + \beta_k x_{ki} + \epsilon_i = \mathbf{x}_i^T \boldsymbol{\beta} + \epsilon_i, \quad i = 1, \ldots, n$$

We can stack these equations for all n equations in the data:

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$$

To estimate this model using least squares, find the coefficient matrix $\boldsymbol{\beta}$ that minimizes the sum of squared residuals:

$$\hat{\boldsymbol{\beta}} = \arg\min_{\boldsymbol{\beta}} = \sum_{i=1}^{n} \left(y_i - \sum_{j=1}^{K} x_{ji}\beta_j \right)^2$$

A more math-y way of saying this using norm notation:

$$\hat{\boldsymbol{\beta}} = \arg\min_{\boldsymbol{\beta}} ||\mathbf{y} - \mathbf{X}\boldsymbol{\beta}||^2$$

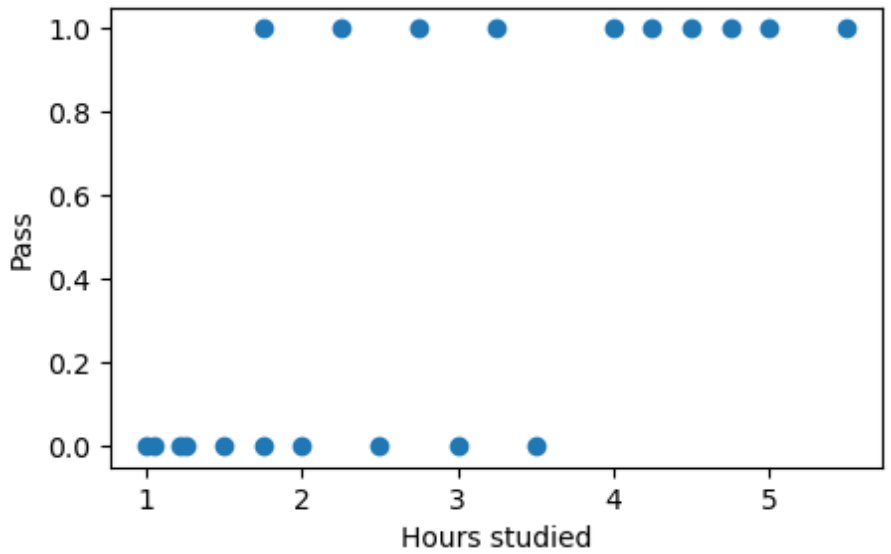
- Just think about the norm notation $||\cdot||$ communicating size in a multidimensional space

Limitations of linear model

We are trying to predict treatment, D_i

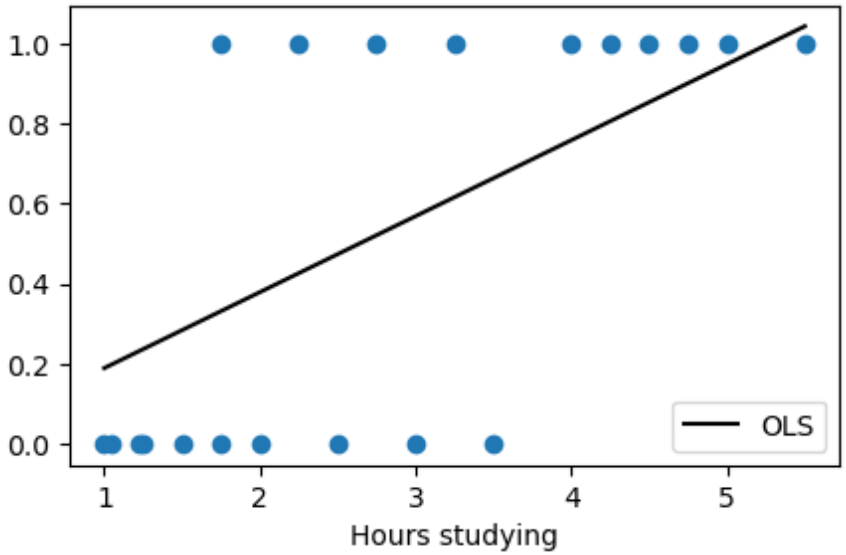
- Always takes on values equal to zero or 1
- Do we really want a linear model?

Consider data on hours studied and passing/failing a test ($D = 1$ implies a student passed)



OLS

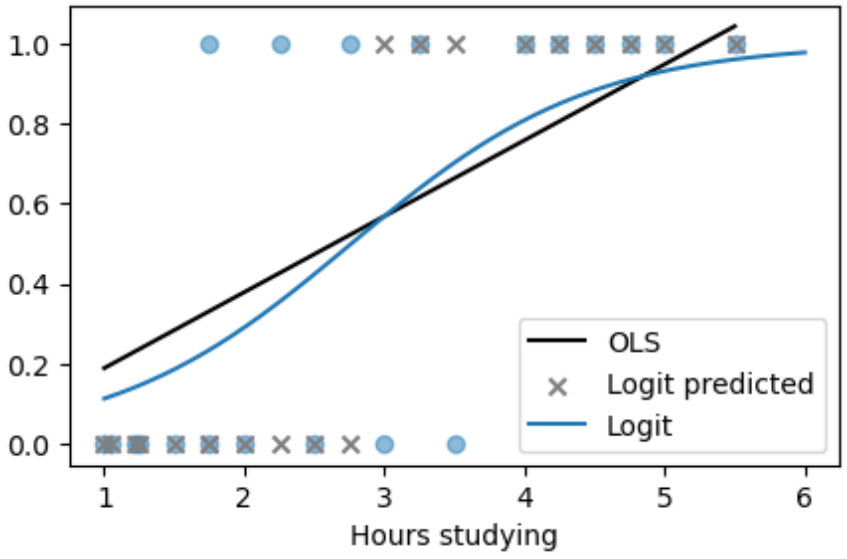
We could estimate using a linear model:



- Line isn't a great fit

Logit

We could estimate using a linear model:



- Logistic line is a better fit
- Less important: output is binary